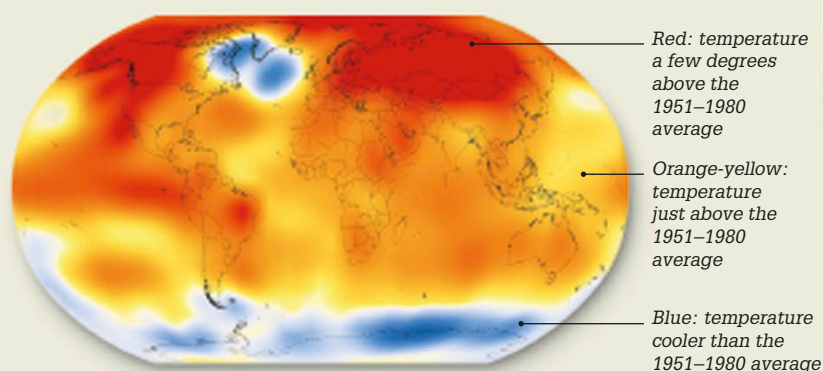


Warming up

Climate monitoring has revealed evidence of average annual temperature increases, with 15 of the 16 warmest years on record occurring since 2001. The map below shows the average annual temperatures of 2015, which were 1.57°F (0.87°C) above the average for the era 1951–1980.



Some radiation from Earth passes through the atmosphere into space.

Some radiation is absorbed by greenhouse gases and radiated back to Earth.

Natural greenhouse gases



Melting away

Melting ice sheets and glaciers are leading to a loss of habitat for animals that live on the ice, as well as rising sea levels, which are a flood threat to low-lying lands. Arctic sea ice in the summer has dropped from 3.02 million sq miles (7.83 million sq km) in 1980 to 1.78 million sq miles (4.63 million sq km) in 2015.

Extreme weather

Global warming is thought to be responsible for an increase in extreme weather events. These include heat waves, heavy rains, and tropical cyclones such as Hurricane Isaac (left), which killed 41 people and caused more than \$2 billion of damage in the southern US, in 2012.



Taking action

Action is being taken at a number of levels to tackle climate change, from international agreements on reducing greenhouse gas emissions to advances in more energy-efficient buildings and technologies, reforestation, and switching away from power generation and vehicles that burn fossil fuels. Individuals can contribute in many ways too, such as:

- ★ Switch off unnecessary electrical items
- ★ Reduce dependence on cars by cycling, walking, or using public transportation
- ★ Use green energy options, such as wind or solar power
- ★ Use energy-saving devices, such as CFL bulbs
- ★ Plant more trees



1988

American professor James Hansen popularized the term “global warming” when reporting to the US Senate that the average global temperature is rising due to the greenhouse effect.

1997

The Kyoto Protocol (an international treaty to bring countries together to reduce global warming) committed developed nations to reduce emissions of key gases responsible for global warming.

2015

NOAA and other bodies reported that 2015 was the hottest year since climate records began in the 19th century.



2016

The Paris Agreement was signed by 180 nations. It aims to hold back global warming to “well below 2°C or 3.6°F” above pre-industrial levels.